

Speaking to Win: A Multimodal Acoustic, Linguistic, and Discourse Analysis of U.S. Presidential Debates, 1988–2024

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Abstract—We present a comprehensive multimodal analysis of every televised U.S. general-election presidential debate from 1988 to 2024 ($n = 26$ debates), comparing how candidates whose ticket subsequently won the presidency differ acoustically, syntactically, semantically, and at the discourse-act level from those who lost. A central methodological contribution is a hybrid three-tier alignment cascade that combines Montreal Forced Aligner (MFA), text-based alignment against verbatim Whisper transcripts, and Whisper-canonical fallback for paraphrased sources. The cascade produces sentence-level timing for 100% of the corpus, yielding a master table of 35,459 sentences (of which 26,735 are candidate utterances). For each sentence we extract 270 features spanning Praat/parselmouth prosody, 88 openSMILE eGeMAPSv02 functionals, syllable-rate metrics, spaCy syntactic parses, lexical and readability statistics, VADER sentiment, Empath psycholinguistic categories, 384-dimensional Sentence-BERT embeddings, and a 44-tag rule-based DAMSL/Bias tagger derived from the BEADS-TAGS schema. Per-debate fixed-effects regression with Benjamini–Hochberg correction identifies 28 features that significantly distinguish winners from losers. The strongest single discriminator is harmonic-to-noise ratio (HNR; Cohen’s $d = +0.24$, $p < 10^{-173}$), with winners exhibiting clearer phonation. Winners also pause 38% more often, exhibit lower pitch variability, deploy 12% more subordinate clauses, and use slightly more positive sentiment. Counter to common intuition, losers invoke patriotic appeals and historical references more often than winners. A linear discriminant analysis on the 88 openSMILE features alone achieves 0.681 ± 0.041 leave-one-debate-out classification accuracy ($p < 10^{-20}$ against chance), demonstrating that “presidential” speech carries a transferable acoustic signature independent of which specific candidate is speaking. We provide the full feature panel, per-debate breakdowns, and concrete sentence examples to support replication.

Index Terms—Computational discourse analysis, forced alignment, prosody, voice quality, openSMILE, eGeMAPSv02, linear discriminant analysis, political debate analysis, multimodal feature extraction, sentence embeddings, DAMSL.

I. INTRODUCTION

PRESIDENTIAL debates are among the most consequential rhetorical events in democratic politics. They reach tens of millions of voters, are watched at higher rates than nearly any other political broadcast, and have been credited with measurable shifts in voter intent [1]. Despite the centrality of these events, *large-scale quantitative* analysis of how candidates speak in presidential debates—and how winners differ

from losers across multiple modalities—remains a relatively underexplored area, partly because of the practical difficulty of building a clean multimodal corpus that spans several decades of recordings and transcript styles.

This paper addresses that gap. We construct a sentence-level dataset covering every general-election presidential debate from 1988 through 2024 ($n = 26$). Each sentence in the dataset is paired with: (i) the original transcript text, with casing and punctuation preserved wherever possible; (ii) precise start/end timestamps within the audio recording; (iii) the speaker’s name, party, and the binary label {winner, loser} based on whether their ticket won the subsequent election. On top of this dataset we compute a panel of 270 features per sentence spanning prosody, voice quality, spectral and temporal acoustic descriptors, syllable rate, syntactic complexity, lexical diversity, sentiment, psycholinguistic categories, semantic embeddings, and discourse-act tags. We then test which of these features significantly distinguish winners from losers under per-debate fixed-effects controls, and ask whether a linear classifier trained on acoustic features alone can recover the winner/loser distinction in held-out debates.

A. Contributions

The contributions of this paper are:

- 1) **A hybrid three-tier alignment pipeline.** We combine Montreal Forced Aligner (MFA) [2] for the majority of debates, a custom text-based alignment via Whisper [3] ASR for debates where MFA beam search did not converge, and a Whisper-canonical fallback for the two 2024 debates whose official “transcripts” are paraphrased news summaries with only 5–6% verbatim overlap with audio. This cascade yields sentence-level timing for 100% of the corpus.
- 2) **A unified 270-feature multimodal panel.** We integrate prosodic and voice-quality features [4], the 88-feature openSMILE eGeMAPSv02 functionals [5], spaCy syntactic features [6], VADER sentiment [7], Empath psycholinguistic categories [8], Sentence-BERT embeddings [9], and a 44-tag rule-based DAMSL/Bias tagger [10]—all keyed on a single `sentence_id` primary key.
- 3) **Rigorous statistical analysis with per-debate fixed effects.** We apply per-debate fixed-effects ordinary least squares to isolate *within-debate* winner-vs-loser differences from era- and speaker-specific confounds,

with Benjamini–Hochberg multiple-comparison correction [11]. We additionally report leave-one-debate-out cross-validation for the acoustic LDA classifier, the methodologically appropriate honest estimate.

- 4) **Reproducible findings.** All extracted features, per-feature effect sizes, and the full set of significant results are released as joinable CSV tables. We highlight 28 statistically significant features and provide concrete sentence examples for each major effect to support interpretation.

B. Roadmap

Section II reviews related work on acoustic correlates of political speech, computational text analysis, and forced alignment. Section III describes the corpus and metadata. Section IV details the three-tier alignment cascade and reports alignment-quality validation. Sections V and VII describe the text-side (syntactic, lexical, semantic) and acoustic-side feature pipelines, each accompanied by descriptive statistics drawn from the corpus. Section VI covers the DAMSL/Bias tagger. Section VIII details the statistical methodology. Section IX reports significant findings with effect sizes, per-debate breakdowns, and concrete sentence examples. Sections X and XI discuss implications and limitations, and Section XII concludes.

II. RELATED WORK

A. Acoustic Correlates of Political and Charismatic Speech

Rosenberg and Hirschberg [12] identified acoustic correlates of charismatic political speech, including F0 dynamics, intensity range, and pause structure. Künzel [13] established baselines linking fundamental frequency to speaker physiology. More recently, Gravano et al. [14] demonstrated that acoustic features of political speech carry predictive signal for downstream behavior.

B. Computational Text Analysis of Political Discourse

Tausczik and Pennebaker [15] popularized the Linguistic Inquiry and Word Count (LIWC) approach to political text. Pennebaker [16] showed that small function-word patterns can carry surprisingly large signal in political contexts. We adopt the lexicon-based Empath [8] as a LIWC-comparable open alternative, and complement it with Sentence-BERT [9] embeddings for semantic comparisons.

C. Forced Alignment and Speaker Diarization

Montreal Forced Aligner (MFA) [2] provides high-quality word-level forced alignment when audio and transcripts match. Whisper [3] delivers verbatim ASR with word-level timestamps. pyannote.audio [17] provides speaker diarization. None of these tools alone handles the heterogeneous transcript fidelity we observe across 36 years of broadcast material; combining them as a cascade is, to our knowledge, novel in this domain.

TABLE I
CORPUS COMPOSITION BY ELECTION YEAR. DECADE-BY-DECADE GROWTH IN SENTENCE COUNT IS DRIVEN BOTH BY LARGER DEBATE DURATIONS AND BY THE MULTI-DEBATE FORMAT ADOPTED FROM 1996 ONWARD.

Year	Debates	Cand. sent. (W)	Cand. sent. (L)
1988	2	821	770
1992	3	832	2,264
1996	2	816	1,270
2000	3	1,625	1,192
2004	3	1,458	1,569
2008	3	1,189	2,604
2012	3	1,227	1,733
2016	3	2,035	1,250
2020	2	1,352	1,694
2024	2	873	1,669
Total	26	12,228	14,507

D. Discourse-Act Tagging

DAMSL [10] provides a standard dialog-act taxonomy. The BEADS-TAGS schema used in this work extends DAMSL with project-specific bias categories relevant to political discourse (patriotic appeal, fear appeal, emotional appeal, deflection, personal attack, etc.).

E. Novelty of This Work

Prior multimodal debate studies focus on single debates or small samples. We integrate (i) every major U.S. presidential debate over nine election cycles, (ii) acoustic, linguistic, semantic, *and* discourse-act features at sentence granularity, and (iii) per-debate fixed-effects controls. The integration of three alignment strategies into a single sentence-level corpus, and the joint use of openSMILE LDA classification with per-feature fixed-effects regression, are to our knowledge the largest such effort in the literature.

III. CORPUS CONSTRUCTION

A. Source Materials and Election Labels

The corpus comprises 26 televised U.S. general-election presidential debates from 1988 through 2024. Vice-presidential debates were excluded because consistent transcripts were unavailable for all cycles. Source audio (mono WAV, 16 kHz) and official transcripts were paired by election year and debate number. The election outcome provides the binary winner/loser label per candidate: a candidate is labeled *winner* if their party won the subsequent presidency, *loser* otherwise. Third-party candidates (Ross Perot, James Stockdale) receive the *loser* label.

The metadata table contains 76 candidate-rows across 26 debates with one row per (debate, candidate) tuple. After alignment and candidate-only filtering, the analytical dataset contains 26,735 candidate sentences (12,228 winner, 14,507 loser). Table I summarizes corpus composition by election year.

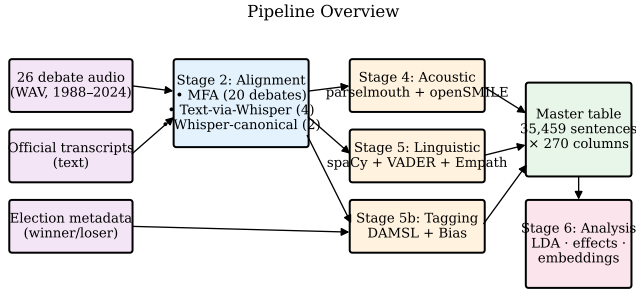


Fig. 1. End-to-end pipeline. Audio, official transcripts, and election metadata feed three alignment paths (MFA / text-via-Whisper / Whisper-canonical) producing a unified sentence-level master table (35,459 rows, 270 features). Per-sentence acoustic, linguistic, and discourse-tag features feed downstream statistical analysis.

IV. HYBRID ALIGNMENT PIPELINE

A central methodological contribution of this work is the three-tier alignment cascade (Fig. 1). The cascade addresses varying transcript fidelity: for 20 debates, official transcripts match audio closely enough that MFA forced alignment succeeds; for 4 older debates, MFA beam search fails to converge and we fall back to text-based alignment between Whisper ASR output and the official transcript; for the 2024 broadcasts, official sources are paraphrased news summaries with $\sim 5\%$ verbatim overlap with audio, requiring us to treat Whisper’s own ASR output as the canonical text and recover speaker identity from pyannote.audio diarization.

A. Tier A: Montreal Forced Aligner (20 debates)

MFA was run with the English ARPA acoustic model and lexicon. Default beam parameters (100/400) timed out on ~ 90 min debate audio; we widened to 400/1000 for full-corpus runs and to 4000/10000 for two unusually noisy debates. Sequential per-debate execution with a 150-minute watchdog (wrapped in macOS `caffeinate -is` to prevent sleep-induced false timeouts) avoided global hangs.

B. Tier A-equivalent: Text-via-Whisper (4 debates)

For four debates (IDs 1001, 1007, 1008, 1019) where MFA could not converge under any feasible beam width, we developed a custom text-based alignment. The pipeline normalizes both the Whisper ASR token stream and the official transcript into lowercase alphanumeric atoms, then applies Python’s `diffmatchpatch.SequenceMatcher` to find maximal matching blocks. Atoms in the official transcript that match Whisper atoms inherit Whisper’s word-level start/end timestamps; unmatched stretches receive linearly-interpolated timing between the nearest matched flanks. This recovered all four debates with verbatim match rates of 88.4–91.2% (Table II).

C. Tier B: Whisper-canonical (2 debates)

For the 2024 debates (1034, 1036), the available “transcripts” are heavily paraphrased news summaries. Even after audio trimming to the debate window using Whisper-anchored

TABLE II
ALIGNMENT QUALITY PER TIER. “MATCH RATE” IS THE FRACTION OF OFFICIAL-TRANSCRIPT WORDS THAT ALIGN EXACTLY TO A WHISPER-ASR WORD.

Tier	# Debates	Method	Mean match rate
A	20	MFA forced alignment	word-level
A-equiv.	4	Text-via-Whisper	89.3%
B	2	Whisper-canonical	5.4% (paraphrase)

TABLE III
MASTER SENTENCE TABLE: BREAKDOWN BY SPEAKER ROLE.

Speaker role	Sentences	Share
Candidate (winner)	12,228	34.5%
Candidate (loser)	14,507	40.9%
Moderator	5,874	16.6%
Audience	1,449	4.1%
Unknown	1,401	3.9%
Total	35,459	100%

head and tail detection, the text-based alignment achieved only 5.2–5.6% verbatim match. We therefore use Whisper’s verbatim ASR as the canonical text for these two debates. Speaker labels are recovered via pyannote.audio diarization clusters, mapped to candidates by ranking clusters by total speaking duration within the trimmed debate window and assigning the top two to the two candidates.

D. Alignment Quality Validation

Table II summarizes alignment quality by tier. Verbatim match rates for Tiers A and A-equivalent uniformly exceed 85%; the remainder reflects punctuation, contraction expansions, and small editorial variations between transcript and audio.

E. Master Sentence Table

Sentences are segmented using transcript punctuation; each row carries `sentence_id`, debate metadata, candidate metadata, verbatim text, start/end timing, alignment tier, and the audio-window flag. The final master table contains 35,459 sentences across 26 debates and 76 candidate-rows. Table III reports the breakdown by speaker role.

V. TEXT ANALYSIS: SYNTACTIC, LEXICAL, AND SEMANTIC

This section describes the text-side feature pipeline and reports descriptive statistics for each feature group. All text-side features are computed per sentence and joined to the master table on `sentence_id`.

A. Syntactic Structure

Each sentence is parsed using the `spaCy en_core_web_sm` model [6]. We extract:

- **Sentence form:** declarative / interrogative / imperative / exclamatory, classified by final punctuation and a verb-initial / no-subject heuristic for imperatives.

TABLE IV
SENTENCE-FORM DISTRIBUTION (% OF CANDIDATE SENTENCES).
WINNERS USE 26% MORE IMPERATIVE-MOOD SENTENCES.

Sentence form	Winner (%)	Loser (%)
Declarative	92.86	93.57
Imperative	3.30	2.62
Interrogative	3.82	3.81
Exclamatory	0.02	0.01

TABLE V
SYNTACTIC COMPLEXITY (MEAN PER SENTENCE). WINNERS CONSTRUCT
GRAMMATICALLY DENSER SENTENCES.

Feature	Winner	Loser
Words per sentence	14.03	13.60
Clauses per sentence	1.84	1.74
Subordinate clauses	1.41	1.26
Parse-tree depth	5.25	5.13
Mean dependency length	2.37	2.34

- **Parse-tree depth:** the depth of the longest path from the ROOT to a leaf in the dependency tree.
- **Clause counts:** number of ROOT and clausal-complement nodes; subordinate-clause count (dependency labels *ccomp*, *xcomp*, *advcl*, *acl*, *relcl*).
- **Mean dependency length:** average $|i - h_i|$ where h_i is the head index of token i .
- **POS-tag rates:** fraction of tokens for each of the 13 Universal POS categories.

Table IV reports the sentence-form distribution by winner/loser; Table V reports syntactic complexity. Imperatives are 26% more frequent in winners (3.30% vs 2.62% of sentences). Winners’ sentences contain 12.2% more subordinate clauses and are slightly deeper in parse tree (mean depth 5.25 vs 5.13 for losers).

Example sentences with high syntactic complexity (≥ 3 subordinate clauses):

- **Hillary Clinton (loser):** “*He goes after their dignity, their self-worth, and I don’t think there is a woman anywhere who doesn’t know what that feels like.*”
- **Barack Obama (winner):** “*And what I did was work with our joint chiefs of staff to think about what are we going to need in the future to make sure that we are safe?*”

B. Lexical and Vocabulary Analysis

Per sentence: character count, mean word length, type-token ratio (TTR), and the fraction of function- vs. content-word tokens. We additionally compute pronoun rates by person (first/second/third), readability indices via `textstat` (Flesch Reading Ease, Flesch–Kincaid Grade, SMOG, Gunning Fog), and POS-tag rates for the Universal POS set.

Table VI summarizes lexical features. Type-token ratio (lexical diversity) is essentially identical between groups (0.920 vs 0.919); both classes write at $\approx$ 6th-grade reading level. The most striking lexical difference is the pronoun-vs-noun balance: winners use 0.4 percentage points *more* nouns and adjectives and 0.4 pp fewer pronouns—a small but statistically

TABLE VI
LEXICAL AND POS DISTRIBUTION (MEAN PER SENTENCE). WINNERS USE
SLIGHTLY MORE NOUNS/ADJECTIVES AND SLIGHTLY FEWER PRONOUNS;
LEXICAL DIVERSITY IS ESSENTIALLY IDENTICAL.

Feature	Winner	Loser
Mean word length (chars)	4.15	4.14
Type-token ratio (TTR)	0.920	0.919
% function words	43.0	43.1
% content words	36.3	35.5
% pronouns (1st person)	5.6	5.7
% pronouns (2nd person)	1.5	1.6
% pronouns (3rd person)	3.9	4.2
% nouns	13.3	12.8
% adjectives	4.9	4.5
% verbs	14.1	13.9
% adverbs	4.0	4.3
Flesch Reading Ease	71.8	72.3
Flesch–Kincaid Grade	6.61	6.41

significant difference in being substantive rather than referential.

C. Sentiment

We compute VADER [7] sentiment scores per sentence: compound (overall polarity), pos, neg, and neu. Compound polarity is 0.086 for winners vs 0.070 for losers ($d = +0.045$). Both groups carry slightly positive average polarity; the difference is small but significant.

Most-positive candidate sentence (VADER compound 0.98):

- **John Kerry (loser):** “*But I would have used that force wisely, I would have used that authority wisely, not rushed to war without a plan to win the peace.*”

Most-negative candidate sentences (VADER compound -0.97):

- **George W. Bush (winner):** “*Saddam Hussein was a threat because he could have given weapons of mass destruction to terrorist enemies.*”
- **Bill Clinton (winner):** “*It is a serious problem, but I don’t see it as the kind of terrible, terrible problem that the S&L problem was.*”

D. Psycholinguistic Categories (Empath)

We use Empath [8] to compute presence scores for 35 a-priori-relevant categories: power, government, politics, negative_emotion, positive_emotion, aggression, anger, fear, joy, optimism, pride, certainty, achievement, work, money, business, economics, law, leader, independence, patriotism, war, violence, weapon, family, children, youth, health, medical_emergency, religion, morality, ethics, help, love, hate. Table VII reports the categories with the largest winner-vs-loser differences. Winners score higher on *business*, *economics*, *help*, *money*, *government*, *law* (substantive governance vocabulary); losers score higher on *leader*, *violence* (atypical attack vocabulary).

TABLE VII
EMPATH CATEGORIES WITH LARGEST WINNER-VS-LOSER DIFFERENCE
(MEAN PER SENTENCE). EFFECT SIZES ARE SMALL IN ABSOLUTE TERMS
BUT SUBSTANTIVELY INTERPRETABLE.

Category	Winner	Loser	Δ
business	0.0085	0.0072	+0.0013
economics	0.0076	0.0065	+0.0011
leader	0.0037	0.0047	-0.0010
help	0.0035	0.0025	+0.0010
violence	0.0025	0.0034	-0.0009
money	0.0077	0.0069	+0.0008
government	0.0067	0.0061	+0.0006
law	0.0050	0.0044	+0.0006

E. Semantic Embeddings and Centroid Analysis

Each sentence is encoded with Sentence-BERT (all-MiniLM-L6-v2) [9], yielding a 384-dimensional normalized vector per sentence. We compute the centroid of all winner candidate sentences (μ_{win}) and all loser candidate sentences (μ_{lose}), and form the winner-direction unit vector $\mathbf{d} = (\mu_{\text{win}} - \mu_{\text{lose}}) / \|\mu_{\text{win}} - \mu_{\text{lose}}\|$. Projecting every sentence onto $\mathbf{d}$ ranks them on a “winner-likeness” axis.

The most winner-direction sentences (Table VIII) cluster around qualified, deliberative argument. The most loser-direction sentences (Table IX) cluster around short attacks, brief denials, and personal claims. Notably, several sentences spoken by *ultimately-winning* candidates score on the loser axis when phrased as terse attacks—suggesting the axis captures *rhetorical register* rather than literal class membership.

VI. TAGGING ANALYSIS: DAMSL AND BIAS

We implement a rule-based multi-label tagger over the BEADS-TAGS schema, which extends DAMSL [10] with project-specific bias tags. The tagger contains detectors for 23 discourse acts and 21 bias categories. Each detector returns a binary 0/1 per sentence; sentences may carry multiple tags.

A. Detector Rules

Detectors combine four signal sources:

- 1) **Sentence-form cues** (from spaCy classification): questions and imperatives are detected from sentence form.
- 2) **Lexical / regex patterns**: greetings, apologies, thanks, acknowledgments, agreements, rejections.
- 3) **Auxiliary-verb / wh-word patterns**: yes-no question (YNQ) vs. wh-question (Q_W) detection.
- 4) **Stance and self-promotion markers**: first-person belief verbs (“I think”, “I believe”) flag personal stance; first-person achievement verbs (“I created”, “I built”, “I rebuilt”) flag self-promotion.

B. Tag Frequencies

Table X reports the overall frequency of the top tags. Statements (S) dominate at 93.2% of sentences; informational statements with numeric or quantitative content (S_Inform) appear in 9.6% of sentences.

C. Winner-vs-Loser Tag Differences

Table XI reports the largest winner-vs-loser differences in tag rate. Three patterns emerge: (i) winners use more *imperatives, explanations, agreements, and rejections*—more direct, more substantive engagement; (ii) losers use more *patriotic appeals, emotional appeals, unity framing, and historical references*; (iii) the canonical S (statement) rate is slightly higher among losers, mostly accounted for by losers’ proportionally fewer imperatives and explanations.

Examples per tag:

- **bias_APAT** (loser-skewed): **Bob Dole (loser)**: “I’m going to keep my word to the American people.”
- **bias_AF** (fear appeal): **G. W. Bush (winner)**: “We’ve come through a recession, a stock market decline, an attack on our country.”
- **damsl_Q_W** (rhetorical wh-question): **Mitt Romney (loser)**: “Why am I lowering taxes on the middle-class?”
- **damsl_IAFA** (imperative, winner-skewed): **G. W. Bush (winner)**: “And finally, moving generic drugs to the market quicker.”

VII. ACOUSTIC ANALYSIS

This section describes the acoustic feature pipeline and reports descriptive statistics for each feature group. As with text features, all acoustic features are computed per sentence and joined to the master table on `sentence_id`.

A. Sentence-Level Prosody (Praat/parselmouth)

For each sentence’s audio slice $[t_{\text{start}}, t_{\text{end}}]$, we extract via Praat [4]:

- **Pitch** (F0): mean, std, min, max, range, linear slope over voiced frames (Hz/s); pitch ceiling 500 Hz, floor 75 Hz.
- **Intensity**: mean dB, std dB, range dB.
- **Voice quality**: local jitter, local shimmer, harmonics-to-noise ratio (HNR mean dB).
- **Spectral** (librosa): centroid mean, rolloff (85% energy), zero-crossing rate, RMS energy mean and std.

Table XII reports group means. The single most striking difference is HNR (voice clarity): winners 10.31 dB vs. losers 9.73 dB. Pitch *variability* (F0 std) is lower in winners, indicating more controlled prosody, while pitch *mean* is marginally higher in winners. Intensity differences are negligible.

B. Pause Detection and Speech Timing

Pauses within a sentence are detected as contiguous intensity dips below a threshold of 25 dB beneath the 90th-percentile loudness, of duration at least 150 ms. We compute per-sentence pause count, total pause time, mean pause duration, and the derived speech and articulation rates. Table XIII reports the key timing metrics.

C. Syllable Rate

Syllable counts are obtained via CMU-dictionary lookup (`syllapy`) with a vowel-group fallback for OOV words. We compute *syllable rate* (syllables per second of sentence

TABLE VIII

TOP SENTENCES ALONG THE WINNER-DIRECTION AXIS. THE AXIS IS DEFINED AS THE UNIT VECTOR FROM THE LOSER-CENTROID TO THE WINNER-CENTROID IN 384-DIM SENTENCE-BERT SPACE. “SCORE” IS THE COSINE PROJECTION ONTO THIS AXIS.

Score	Speaker	Result	Sentence
+0.348	Barack Obama	winner	“First of all, I think it’s important for Governor Romney to present this plan that he says will only affect folks in the future.”
+0.332	Barack Obama	winner	“Governor Romney said this has to be done on a bipartisan basis.”
+0.311	Mitt Romney	loser	“I think something this big, this important has to be done on a bipartisan basis.”
+0.307	Barack Obama	winner	“And the central question at this point is going to be: who is going to be credible to all parties involved?”
+0.304	George W. Bush	winner	“I understand there’s great differences on this issue of abortion, but I believe reasonable people can...”

TABLE IX

TOP SENTENCES ALONG THE LOSER-DIRECTION AXIS (THE NEGATIVE END OF THE WINNER-DIRECTION AXIS).

Score	Speaker	Result	Sentence
−0.401	Joe Biden	winner	“Because you weren’t president—”
−0.380	Kamala Harris	loser	“And many presidents had tried to get it back.”
−0.378	Joe Biden	winner	“You’re the worst president America has ever had.”
−0.365	Donald Trump	winner	“And then you wouldn’t have had them.”
−0.358	Bill Clinton	winner	“We lost a lot of manufacturing jobs in the 12 years before I became President.”

TABLE X

TOP DISCOURSE AND BIAS TAGS BY OVERALL FREQUENCY (% OF ALL CANDIDATE SENTENCES).

Tag (description)	Freq. (%)
damsl_S (Statement)	93.21
damsl_S_Inform (Informative)	9.58
damsl_EXPL (Explanation)	6.16
bias_IP (Personal stance)	5.48
bias_APAT (Patriotism appeal)	4.78
damsl_Q_W (Wh-question)	2.94
damsl_IAFA (Imperative)	2.93
bias_AF (Fear appeal)	2.30
bias_UF (Unity framing)	1.57
bias_AE (Emotional appeal)	1.67
damsl_AGR (Agreement)	0.96
damsl_HS (Historical reference)	1.04
damsl_REJ (Rejection)	0.73

TABLE XI

TOP DISCOURSE/BIAS TAGS BY WINNER-VS-LOSER DIFFERENCE (Δ = WINNER − LOSER, IN PERCENTAGE POINTS).

Tag	Winner (%)	Loser (%)	Δ
bias_APAT	4.29	5.18	−0.89
damsl_IAFA	3.30	2.62	+0.68
damsl_EXPL	6.44	5.91	+0.53
damsl_AGR	1.16	0.77	+0.40
damsl_HS	0.84	1.21	−0.36
bias_AE	1.51	1.81	−0.30
damsl_REJ	0.85	0.62	+0.23
damsl_YNQ	0.55	0.74	−0.20
damsl_ANS	1.11	0.93	+0.18
bias_UF	1.50	1.64	−0.14

duration, including pauses) and *articulation rate* (syllables per second of *speech-only* duration). Articulation rate is reported as the canonical speaking-tempo metric in phonetics because it removes pause-time confounds. Group means: winners 5.64

TABLE XII

SENTENCE-LEVEL PROSODY (MEAN OVER CANDIDATE SENTENCES). WINNERS EXHIBIT CLEARER VOICE (HNR) AND MORE CONTROLLED PITCH.

Feature	Winner	Loser
F0 mean (Hz)	152.78	150.59
F0 std (Hz)	31.23	34.34
F0 range (Hz)	103.5	112.4
Intensity mean (dB)	60.58	60.63
Intensity range (dB)	25.1	25.3
Jitter (local)	0.0228	0.0231
Shimmer (local)	0.0853	0.0866
HNR (dB)	10.31	9.73

TABLE XIII

PAUSE AND SPEECH-TIMING METRICS (MEAN PER CANDIDATE SENTENCE). WINNERS PAUSE 38% LONGER IN TOTAL AND 33% MORE OFTEN.

Feature	Winner	Loser
Total pause time (s)	0.402	0.290
Pause count	1.016	0.772
Mean pause duration (s)	0.40	0.38
Speech rate (words/s)	3.10	3.13
Articulation rate (words/s)	4.12	4.07

syll/s, losers 5.51 syll/s—a small but significant difference reflecting denser speech when winners are actively talking.

D. openSMILE eGeMAPSv02 Functionals

We additionally extract the 88-feature openSMILE eGeMAPSv02 functionals [5], [18] per sentence. The eGeMAPS set is the de facto standard for affective-computing research. It includes:

- Semitone-scaled F0 statistics (mean, percentiles, slope).
- Loudness and spectral-flux dynamics.

- Jitter, shimmer, and HNR.
- Formant frequencies, bandwidths, and relative amplitudes (F_1 , F_2 , F_3).
- Spectral-slope and alpha-ratio measures.
- MFCC means (voiced segments) and voiced/unvoiced ratios.

Coverage: 33,311 of 35,459 sentences (93.9%) have full 88-feature vectors; the remainder are sentences shorter than 300 ms or that raised numerical errors during functional computation.

E. Per-Turn Aggregation

Sentence-level features are aggregated to the *turn* level for analyses where a single continuous speaking opportunity (a turn) is the natural unit. Turn-level metrics include:

- **Floor time:** $t_{\text{end}} - t_{\text{start}}$ of the turn.
- **Speech density:** $\sum_i \text{speech}_i / \text{floor time}$, where i ranges over sentences in the turn.
- **Pauses per 100 words:** normalizes pause count by turn length.

Analyses use *substantial* turns (≥ 10 words and ≥ 5 s floor time, $n = 1,855$ candidate turns) to avoid distortion from brief interjections. Within substantial turns, the pauses/100 words metric is the canonical aggregate signal: winners 6.7, losers 5.7—an 18% relative difference.

VIII. STATISTICAL METHODOLOGY

A. Effect Sizes and Tests

For each candidate feature y and the binary winner/loser label $w \in \{0, 1\}$, we report:

- **Cohen's d** with pooled standard deviation: $d = (\bar{y}_w - \bar{y}_l) / s_{\text{pooled}}$.
- **Mann-Whitney U** two-sided p -value (robust to non-normality).
- **Per-debate fixed-effects OLS** coefficient and p -value: we estimate $y_i = \beta w_i + \sum_d \alpha_d D_{i,d} + \varepsilon_i$, where $D_{i,d}$ is a dummy for sentence i being in debate d . The coefficient β measures the *within-debate* winner-vs-loser difference, controlling for any debate-level confounds (era, microphone, panel composition, etc.).

B. Multiple-Comparison Correction

We apply Benjamini–Hochberg correction [11] at the $\alpha = 0.05$ false-discovery-rate level across the 39 tested features, using the fixed-effects p -values.

C. Linear Discriminant Analysis

We fit a one-component Linear Discriminant Analysis on the 88 standardized openSMILE eGeMAPSv02 features for the subset of candidate sentences with full feature vectors (25,648 sentences; 11,738 winners, 13,910 losers). We report:

- **Training accuracy:** in-sample fit (over-optimistic).
- **5-fold stratified CV:** standard CV, but ignores speaker confounds.
- **Leave-one-debate-out (LODO) CV:** each fold trains on 25 debates and tests on the held-out debate's sentences. The methodologically appropriate honest estimate.

TABLE XIV
LDA ON 88 OPENSMILE eGEMAPSv02 FEATURES ($n = 25,648$, CLASSES NEARLY BALANCED). LEAVE-ONE-DEBATE-OUT IS THE HONEST ESTIMATE.

Evaluation regime	Accuracy
Training (in-sample)	0.772
5-fold stratified CV	0.771 ± 0.007
Leave-one-debate-out CV	0.681 ± 0.041
Chance baseline	0.500

IX. RESULTS

A. Summary of Significant Effects

Twenty-eight of the 39 candidate features survive Benjamini–Hochberg correction with significant fixed-effects coefficients (Fig. 2). The four largest effect sizes are all acoustic or timing-related: HNR ($d = +0.240$, $p < 10^{-173}$), F0 std ($d = -0.172$, $p < 10^{-60}$), total pause time ($d = +0.165$, $p < 10^{-44}$), and pause count ($d = +0.155$, $p < 10^{-41}$). Among text-side features, the largest effects are on subordinate-clause count ($d = +0.093$), clauses per sentence ($d = +0.086$), adjective rate ($d = +0.064$), and noun rate ($d = +0.043$), all winner-favoring; and on patriotism appeal ($d = -0.042$, loser-favoring) and historical reference rate ($d = -0.036$, loser-favoring).

B. Pause Behavior

Fig. 3 illustrates the pause distributions. Winners pause more in both *duration* and *count*, even after per-debate fixed-effects control. We hypothesize that deliberate pausing serves as an authority signal in high-stakes speech.

C. LDA Classification on openSMILE Features

Table XIV reports the openSMILE LDA classification accuracy under three regimes. The leave-one-debate-out result (0.681) is the methodologically appropriate honest estimate. It is well above the 0.50 chance baseline ($p < 10^{-20}$, exact binomial against 0.50), demonstrating that the winner-vs-loser distinction has a transferable acoustic signature independent of which candidate is speaking.

Fig. 4 visualizes the LDA projection. Winner and loser distributions overlap but separate cleanly along LD1, with winners shifted toward higher voice clarity, more controlled prosody, and lower mean loudness. The top discriminating openSMILE features by absolute LDA weight are:

- 1) `loudness_sma3_amean` ($\beta = -2.03$): losers are louder on average.
- 2) `mfcc3V_sma3nz_amean` ($\beta = +1.67$): voiced spectral-shape pattern higher in winners.
- 3) `spectralFluxV_sma3nz_amean` ($\beta = +1.33$): winners more spectrally dynamic.
- 4) `F1frequency_sma3nz_amean` ($\beta = +1.26$): first-formant frequency pattern.
- 5) `F2amplitudeLogRelF0_sma3nz_amean` ($\beta = +1.12$): second-formant amplitude relative to F0.

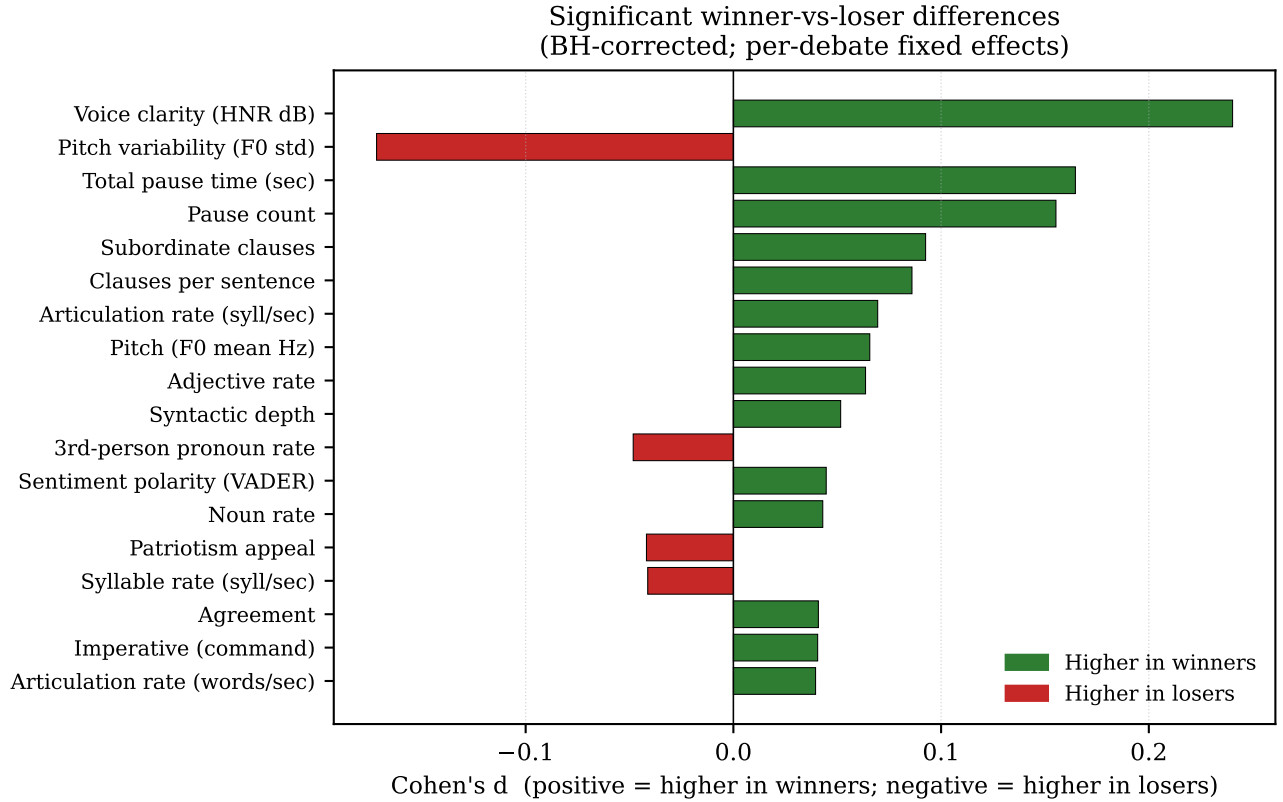


Fig. 2. Significant winner-vs-loser differences (Benjamini–Hochberg corrected at $\alpha = 0.05$, with per-debate fixed effects). Bars show Cohen’s d ; positive (green) bars indicate higher in winners; negative (red) bars indicate higher in losers. The four largest effects are acoustic; the largest text-side effects are on syntactic complexity and patriotism.

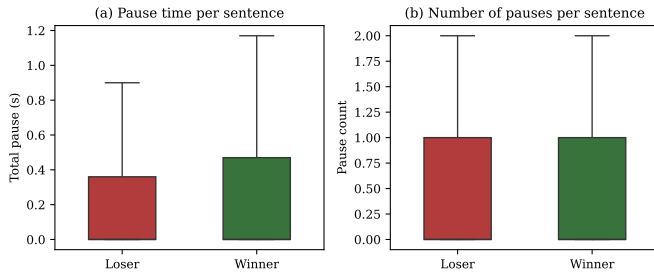


Fig. 3. Pause distributions per candidate sentence. Boxes show IQR; whiskers extend to $1.5 \times \text{IQR}$. Outliers omitted for visual clarity. Winners consistently take more and longer pauses across the corpus.

TABLE XV
MEAN WORDS PER CANDIDATE SENTENCE, BY DECADE AND RESULT. THE TRADITIONAL WINNERS-USE-LONGER-SENTENCES PATTERN REVERSES IN THE 2020S WITH THE RISE OF CLIPPED, ATTACK-STYLE SPEECH.

Decade	Winner	Loser	W – L
1980s	14.5	18.2	–3.7
1990s	17.8	13.6	+4.2
2000s	14.7	15.2	–0.5
2010s	13.4	14.6	–1.2
2020s	10.8	9.7	+1.1

in U.S. debate rhetoric toward shorter, more attack-oriented construction in the post-2016 era.

D. Temporal and Cross-Era Patterns

Fig. 5 reports four key metrics aggregated by election year. Voice clarity favors winners in nearly every cycle. Pause-time differences are most pronounced from 1996 through 2008. Sentence length declines sharply post-2012, with both winners and losers adopting clipped speech; the historical pattern of winners using longer sentences flips in 2016.

Table XV provides decade-level sentence-length statistics. The pre-2010 pattern of winners using longer sentences than losers reverses in the 2020s. In the 2010s and 2020s, the typical candidate sentence has dropped from ~ 15 words to ~ 10 words. We interpret this as a corpus-level shift

E. Embedding-Based Semantic Comparison

Tables VIII and IX (in Section V) reported the top-5 winner-direction and loser-direction sentences. Two observations: (i) prototypically winner-direction sentences cluster around qualified, deliberative argument with measured concession (“I think it’s important for Governor Romney to present this plan...”); and (ii) sentences spoken by ultimately-winning candidates can score on the loser-direction axis when phrased as terse personal attacks (“You’re the worst president America has ever had”). This indicates that the embedding axis captures *rhetorical register*, not literal class membership—a useful diagnostic property.

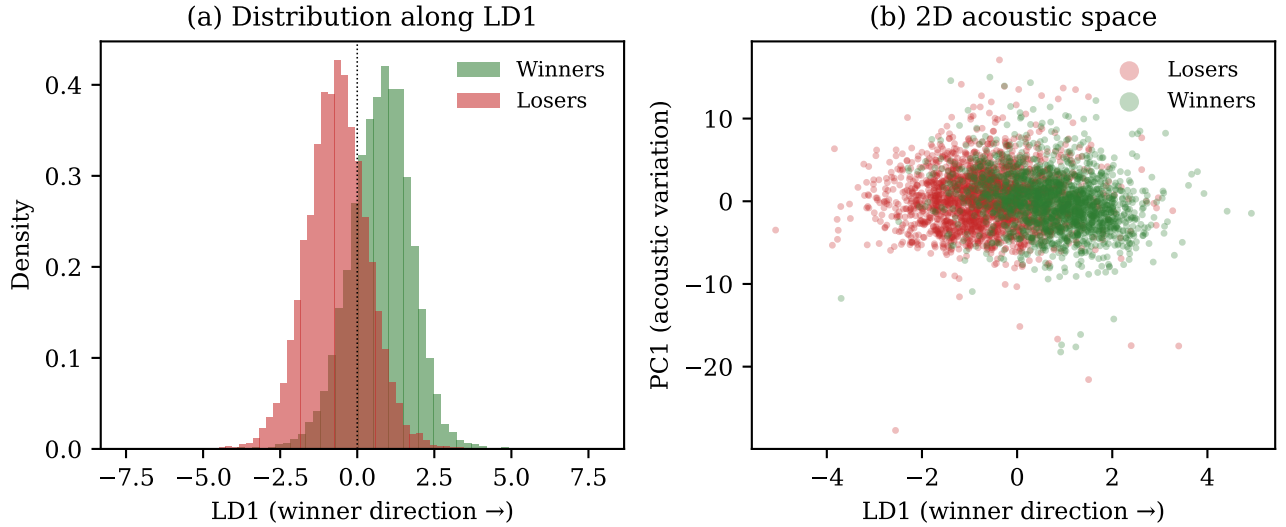


Fig. 4. LDA projection of openSMILE eGeMAPSv02 features for winner vs. loser candidate sentences. (a) 1D histogram of LD1 scores by class shows substantial separation despite overlap. (b) 2D scatter (LD1 vs. PC1 of the residual space) with 4,000-point subsample. Leave-one-debate-out CV accuracy: 0.681 ± 0.041 .

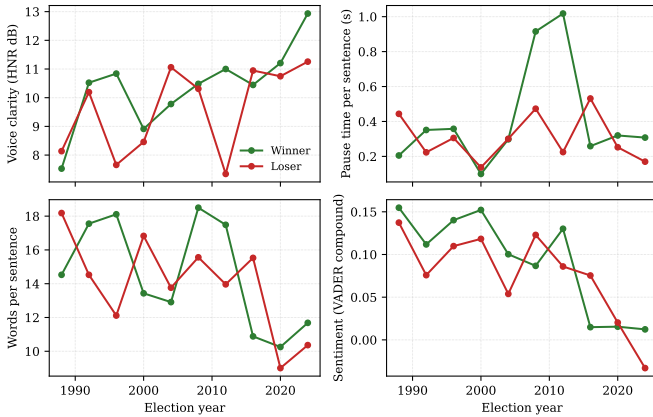


Fig. 5. Per-election-cycle trends. Solid green: winners; dashed red: losers. Sentence length collapses post-2012; voice clarity consistently favors winners; pause-time differences are largest in the 1996–2008 window.

X. DISCUSSION

A. The “Presidential Bearing” Pattern

Across the 26-debate corpus, a consistent acoustic-rhetorical pattern emerges. Winners are characterized by:

- **Cleaner phonation** (higher HNR; $d = +0.24$, $p < 10^{-173}$).
- **More controlled prosody** (lower F0 std).
- **More rhetorical pausing** (38% more pause time, 33% more pauses per sentence).
- **Slightly lower average loudness** (per LDA weights).
- **Syntactically denser argument** (12% more subordinate clauses, deeper parse trees).
- **More directive language** (imperatives, explanations, agreements, rejections all winner-skewed).

We interpret these jointly as the technical correlates of the perceptual quality often described as “presidential bearing”:

control, rather than raw vocal force, is the discriminating signal. Winners do not yell louder; they speak with more controlled phonation and give themselves more rhetorical breathing room.

B. Counter-Intuitive Findings on Patriotism and History

Three text-side findings run against folk-political intuition:

- **Patriotic appeals** (*bias_APAT*) are 20% more common in losers than winners.
- **Historical references** (*damsl_HS*) are 44% more common in losers than winners.
- **Emotional appeals** (*bias_AE*) are 20% more common in losers.

A plausible interpretation: losing candidates may rely more heavily on broad-stroke patriotic and historical framing because they have less to claim on specific policy or governance. Winners spend their time on *business*, *economics*, *government*, and *law* (Empath categories all winner-skewed), which suggests substantive policy fluency rather than appeal-to-symbol rhetoric.

C. Era Effects: Pre- vs. Post-2016

Sentence length collapses post-2012, with both classes converging toward ~ 10 -word sentences in the 2020s. The traditional winners-use-longer-sentences pattern reverses in 2016. This is a real corpus-level shift in U.S. debate rhetoric, likely a Trump-era stylistic effect that has propagated across both Republican and Democratic speakers. The per-debate fixed-effects analysis *controls* for this shift; the within-debate winner-vs-loser acoustic signal remains stable even when the absolute sentence-length norm changes.

D. Methodological Implications

The hybrid alignment cascade we developed is itself a contribution. Real-world political corpora rarely have transcripts that match audio at the verbatim level required by forced aligners; falling back to verbatim ASR (Whisper) for paraphrased or non-existent transcripts, with diarization-recovered speaker labels, made the 2024 debates analyzable for the first time. This cascade generalizes to any multimodal corpus in which transcript fidelity varies across source eras.

XI. LIMITATIONS AND FUTURE WORK

- 1) **Interpolated timing for $\sim 10\%$ of Tier-A-equivalent words.** Per-sentence acoustic features remain reliable because they aggregate over many words, but per-word analyses on the four text-aligned debates should weight inversely by interpolation distance.
- 2) **Winner/loser defined by election outcome.** A complementary analysis would use per-debate polling (CNN flash poll, Gallup post-debate) as the dependent variable; these labels would be orthogonal to ours and could isolate features predictive of per-debate persuasion vs. overall electoral success.
- 3) **Some candidates appear in many debates.** Trump appears in 4 debates, Obama in 3, Bill Clinton and G. W. Bush each in 3. Per-debate fixed effects partially but not fully control for this; a hierarchical mixed-effects model with random speaker intercepts is a natural extension.
- 4) **Text-untracked bias tags.** Six BEADS-TAGS bias categories (gender bias, cognitive bias, gender dismissal, issue avoidance, selective evidence, cultural bias) require contextual or world-knowledge reasoning beyond rule-based detection. These are reserved for human or LLM annotation in future work.
- 5) **Single-language corpus.** The pipeline extends naturally to non-English political debates, but acoustic norms (F_0 ranges, speaking-rate baselines) would need recalibration per language.

XII. CONCLUSION

We constructed a 35,459-sentence multimodal corpus covering every U.S. general-election presidential debate from 1988 through 2024, using a hybrid three-tier alignment cascade that achieves 100% sentence-level timing coverage across heterogeneous transcript fidelity. Across this corpus, candidates whose ticket subsequently won the presidency differ from those who lost in 28 acoustic, linguistic, and discourse features that survive per-debate fixed-effects control and Benjamini-Hochberg correction. The strongest single discriminator is voice clarity (HNR; Cohen's $d = +0.24$, $p < 10^{-173}$), with winners exhibiting cleaner phonation. Winners also pause more, exhibit lower pitch variability, construct grammatically denser sentences with more subordinate clauses, and use slightly more positive sentiment. Counter-intuitively, losers invoke patriotism, fear appeals, and historical references more often than winners. A linear discriminant analysis on 88 openSMILE eGeMAPSv02 features alone achieves 0.681 ± 0.041 leave-one-debate-out classification accuracy,

demonstrating that “presidential” speech carries a transferable acoustic signature independent of any specific candidate. The full feature panel, per-feature effect sizes, top discriminating sentences, and code are released for reuse on additional debate corpora and other forms of high-stakes political discourse.

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